METRIC

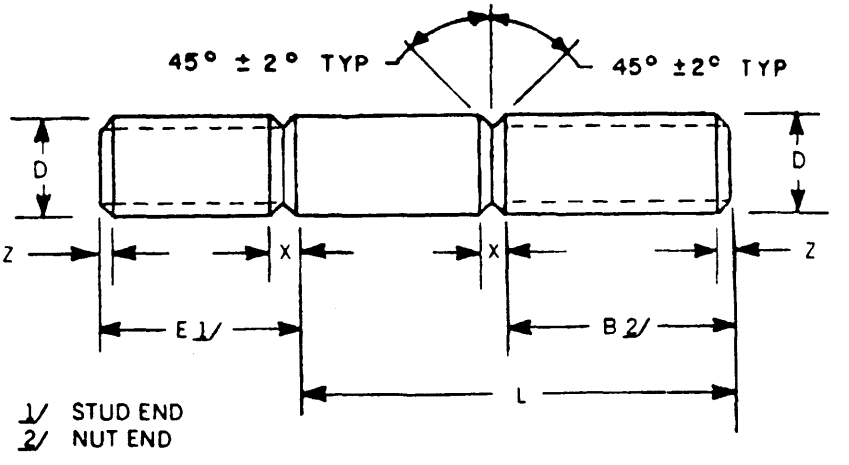
DOD-S-63543/1
19 February 1982

MILITARY SPECIFICATION SHEET

STUD, PLAIN, METRIC
(1.25 DIA. AND 2 DIA. ENGAGEMENT)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the studs described herein shall consist of this specification and the latest issue of specification DOD-S-63543.



NOTES:
1. Unless otherwise specified, all dimensions are in millimeters.
2. Unthreaded portion of stud shall be the Major diameter of the thread as specified in FED-STD-H28/21 Table: External Threads-Limiting Dimension MJ-Profile.

FIGURE 1. Stud, plain.

FSC 5307

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REQUIREMENTS:

1. Material: Steel, alloy, finish-cadmium plate, Code: A or steel, alloy, finish-zinc plate, Code: Z.
   Hardness: 26 to 32 HRC.
   Manufacturing specification: AMS 7456 except thread form.

2. Material: Brass, Code: B.
   Finish: Black chemical.
   Hardness: 65 HRB min.

3. Material: CRES, Code: C.
   Finish: Passivate.
   Hardness: 140 HRB min.

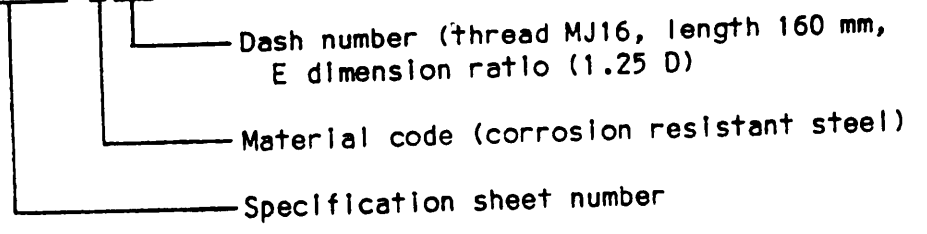
4. Material: Nickel-copper alloy, Code: D.
   Finish: None.
   Hardness: 80 HRB min.

5. Surface texture: ANSI B 46.1, unless otherwise specified, surfaces to be 3.2 µm.

6. Thread profile and tolerance: MJ - 4h 6h.

7. Part numbers: Part numbers consists of the letter "D" the basic number of this specification sheet, plus the material code letter, plus the stud size dash number. Do not use unassigned part numbers.

EXAMPLE: D63543/1-C199



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TABLE I. 1.25 Diameter engagements.

D Thread		MJ4	MJ5	MJ6	MJ8	MJ10	MJ12	MJ16
Pitch		0.7	0.8	1	1	1.25	1.25	1.5
B	1/	14	16	18	22	26	30	38
	2/	20	22	24	28	32	36	44
	3/	--	--	--	--	45	49	57
E		5	6.25	7.5	10	12.5	15	20
X		1.75	2	2.5	2.5	3.2	3.2	3.8
Z		1	1.2	1.5	1.5	1.8	1.8	2.2

Length	Tol	Dash numbers						
16	± 0.25	--	105	119	133	--	--	
20	± 0.25	100	106	120	134	149	164	
25	± 0.25	101	107	121	135	150	165	
30	± 0.25	102	108	122	136	151	166	181
35	± 0.25	103	109	123	137	152	167	182
40	± 0.25	104	110	124	138	153	168	183
45	± 0.25		111	125	139	154	169	184
50	± 0.25		112	126	140	155	170	185
55	± 1.0		113	127	141	156	171	186
60	± 1.0		114	128	142	157	172	187
65	± 1.0		115	129	143	158	173	188
70	± 1.0		116	130	144	159	174	189
75	± 1.0		117	131	145	160	175	190
80	± 1.0		118	132	146	161	176	191
90	± 1.0		--	--	147	162	177	192
100	± 1.0				148	163	178	193
110	± 1.5				--	--	179	194
120	± 1.5				--	--	180	195
130	± 1.5							196
140	± 1.5							197
150	± 1.5							198

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TABLE I. 1.25 Diameter engagements. - Continued

D Thread		MJ20	MJ24	MJ30	MJ36	MJ42	MJ48
Pitch		1.5	2	2	2	2	2
B	1/	46	54	66	78	90	102
	2/	52	60	72	84	96	108
	3/	65	73	85	97	109	121
E		25	30	37.5	45	52.5	60
X		3.8	5	5	7.5	7.5	7.5
Z		2.2	3	3	4.5	4.5	4.5

Length	Tol	Dash numbers					
16	± 0.25						
20	± 0.25						
25	± 0.25						
30	± 0.25	--	--				
35	± 0.25	--	--				
40	± 0.25	200	221				
45	± 0.25	201	222				
50	± 0.25	202	223				
55	± 1.0	203	224				
60	± 1.0	204	225				
65	± 1.0	205	226				
70	± 1.0	206	227				
75	± 1.0	207	228	--			
80	± 1.0	208	229	--			
90	± 1.0	209	230	242			
100	± 1.0	210	231	243	259	--	
110	± 1.5	211	232	244	260	--	
120	± 1.5	212	233	245	261	278	
130	± 1.5	213	234	246	262	279	297
140	± 1.5	214	235	247	263	280	298
150	± 1.5	215	236	248	264	281	299

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TABLE I. 1.25 Diameter engagement. - Continued

D Thread		MJ4	MJ5	MJ6	MJ8	MJ10	MJ12	MJ16
Pitch		0.7	0.8	1	1	1.25	1.25	1.5
B	1/	14	16	18	22	26	30	38
	2/	20	22	24	28	32	36	44
	3/	--	--	--	--	45	49	57
E		5	6.25	7.5	10	12.5	15	20
X		1.75	2	2.5	2.5	3.2	3.2	3.8
Z		1	1.2	1.5	1.5	1.8	1.8	2.2

Length	Tol	Dash numbers						
160	± 1.5							199
170	± 1.5							--
180	± 1.5							--
190	± 1.5							
200	± 1.5							
220	± 1.5							
240	± 1.5							
260	± 1.5							
280	± 1.5							
300	± 1.5							
320	± 1.5							
340	± 1.5							
360	± 1.5							
380	± 1.5							
400	± 1.5							

1/ Less than 125

2/ 125 to 200

3/ More than 200

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TABLE I. 1.25 Diameter engagement. - Continued

D Thread		MJ20	MJ24	MJ30	MJ36	MJ42	MJ48
Pitch		1.5	2	2	2	2	2
B	1/	46	54	66	78	90	102
	2/	52	60	72	84	96	108
	3/	65	73	85	97	109	121
E		25	30	37.5	45	52.5	60
X		3.8	5	5	7.5	7.5	7.5
Z		2.2	3	3	4.5	4.5	4.5
Length	Tol	Dash numbers					
160	± 1.5	216	237	249	265	282	300
170	± 1.5	217	238	250	266	283	301
180	± 1.5	218	239	251	267	284	302
190	± 1.5	219	240	252	268	285	303
200	± 1.5	220	241	253	269	286	304
220	± 1.5	--	--	254	270	287	305
240	± 1.5			255	271	288	306
260	± 1.5			256	272	289	307
280	± 1.5			257	273	290	308
300	± 1.5			258	274	291	309
320	± 1.5			--	275	292	310
340	± 1.5			--	276	293	311
360	± 1.5				277	294	312
380	± 1.5				--	295	313
400	± 1.5				--	296	314

1/ Less than 125

2/ 125 to 200

3/ More than 200

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TABLE II. 2 Diameter engagement.

D Thread		MJ4	MJ5	MJ6	MJ8	MJ10	MJ12	MJ16	MJ20	MJ24
Pitch		0.7	0.8	1	1.25	1.25	1.5	1.5	1.5	2
B	1/	14	16	18	22	26	30	38	46	54
	2/	20	22	24	28	32	36	44	52	60
	3/	--	--	--	--	45	49	57	65	73
E		8	10	12	16	20	24	32	40	48
X		1.75	2	2.5	2.5	3.2	3.2	3.8	3.8	5
Z		1	1.2	1.5	1.5	1.8	1.8	2.2	2.2	3
Length	Tol	Dash numbers								
16	± 0.25	--	--	--						
20	± 0.25	400	--	--						
25	± 0.25	401	405	411						
30	± 0.25	402	406	412	419	--	--			
35	± 0.25	403	407	413	420	430	--			
40	± 0.25	404	408	414	421	431	442			
45	± 0.25		409	415	422	432	443	--		
50	± 0.25		410	416	423	433	444	455		
55	± 1.0		--	417	424	434	445	456		
60	± 1.0			418	425	435	446	456	470	--
65	± 1.0			--	426	436	447	457	471	--
70	± 1.0			--	427	437	448	458	472	487
75	± 1.0				428	438	449	459	473	488
80	± 1.0				429	439	450	460	474	489
90	± 1.0				--	440	451	461	475	490

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TABLE II. 2 Diameter engagement. - Continued

D Thread		MJ4	MJ5	MJ6	MJ8	MJ10	MJ12	MJ16	MJ20	MJ24
Pitch		0.7	0.8	1	1.25	1.25	1.5	1.5	1.5	2
B	1/	14	16	18	22	26	30	38	46	54
	2/	20	22	24	28	32	36	44	52	60
	3/	--	--	--	--	45	49	57	65	73
E		8	10	12	16	20	24	32	40	48
X		1.75	2	2.5	2.5	3.2	3.2	3.8	3.8	5
Z		1	1.2	1.5	1.5	1.8	1.8	2.2	2.2	3
Length	Tol	Dash numbers								
100	± 1.0					441	452	462	476	491
110	± 1.5					--	453	463	477	492
120	± 1.5					--	454	464	478	493
130	± 1.5							465	479	494
140	± 1.5							466	480	495
150	± 1.5							467	481	496
160	± 1.5							468	482	497
170	± 1.5							469	483	498
180	± 1.5							--	484	499
190	± 1.5								485	500
200	± 1.5								486	501

1/ Less than 125

2/ 125 to 200

3/ More than 200

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